

An All-Period Nonlattice Suspension Determinant with Exact Clock-Sector Separation

Route-A structural certificate C130

Abstract

We freeze the full binary shift with roof values 1 and $\sqrt{2}$. Its two-state bivariate matrix has determinant $1 - u - v$; Laplace specialization gives the entire exponential polynomial $1 - e^{-s} - e^{-\sqrt{2}s}$. The same source owns an all-period primitive dynamical Euler/trace identity. Irrationality separates symbol-count clock sectors, while the rational roof $(1, 2)$ restores collisions and vertical periodicity. These are source-dynamical statements, not a target divisor or arithmetic correspondence.

1 Frozen mixing suspension

Let $B = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ and let Σ_B be the two-sided full binary shift. Since B is positive, the shift is mixing. Freeze $\tau(0) = 1$, $\tau(1) = \sqrt{2}$ and form $X^\tau = (\Sigma_B \times \mathbb{R}) / ((x, t + \tau(x)) \sim (\sigma x, t))$. For a closed word w , write $N_j(w)$ for its symbol populations and

$$\ell(w) = N_0(w) + \sqrt{2}N_1(w).$$

2 Bivariate owner and primitive identity

Destination-symbol weights give

$$M(u, v) = B \operatorname{diag}(u, v) = \begin{pmatrix} u & v \\ u & v \end{pmatrix}, \quad \Delta(u, v) = \det(I - M) = 1 - u - v.$$

The eigenvalues are $u + v, 0$, hence for every $n \geq 1$,

$$\operatorname{Tr} M^n = (u + v)^n = \sum_{k=0}^n \binom{n}{k} u^{n-k} v^k. \quad (1)$$

The trace sums all rooted binary closed words. For $|u| + |v| < 1$,

$$-\log \Delta(u, v) = \sum_{n \geq 1} \frac{\operatorname{Tr} M(u, v)^n}{n}.$$

Decomposing each word uniquely as a repetition of a primitive cyclic word γ and regrouping this absolutely convergent series gives

$$\Delta(u, v) = \prod_{[\gamma]} (1 - u^{N_0(\gamma)} v^{N_1(\gamma)}). \quad (2)$$

Thus with $u = e^{-s}$ and $v = e^{-\sqrt{2}s}$,

$$d_\tau(s) = 1 - e^{-s} - e^{-\sqrt{2}s} = \prod_{[\gamma]} (1 - e^{-s\ell(\gamma)}). \quad (3)$$

Equation (2) also holds formally in the total-degree completion, since each degree sees finitely many cycles. Let $h > 0$ solve $e^{-h} + e^{-\sqrt{2}h} = 1$. The product (3) converges absolutely for $\operatorname{Re} s > h$. The explicit left side is entire and its reciprocal is meromorphic; the product is not asserted to converge globally.

3 Irrational clock and rational control

If $a + b\sqrt{2} = c + d\sqrt{2}$ for integers, irrationality gives $(a, b) = (c, d)$. Thus different population vectors never collide in suspension time. This is not orbit injectivity: the distinct primitive necklaces 000111 and 001011 both have vector $(3, 3)$ and roof $3 + 3\sqrt{2}$.

Change only the roof to $(1, 2)$. Then

$$d_{\text{rat}}(s) = 1 - e^{-s} - e^{-2s} = 1 - q - q^2, \quad q = e^{-s}.$$

The second repetition of $[0]$, with counts $(2, 0)$, collides at time 2 with primitive fixed orbit $[1]$, with counts $(0, 1)$. Also $d_{\text{rat}}(s + 2\pi i) = d_{\text{rat}}(s)$: lattice periodicity is restored.

4 Replay and Route-A boundary

Exact replay gives

n	1	2	3	4	5	6	7	8	9	10
rooted	2	4	8	16	32	64	128	256	512	1024
primitive	2	1	2	3	6	9	18	30	56	99
sectors	2	3	4	5	6	7	8	9	10	11

Totals are 2046 rooted words, 226 primitive cycles, and 65 sectors. The independent standard-library checker, fresh symbolic reconstruction, byte replay, and all 31 hostile mutations pass.

The strict tuple is (A1_WEAK, A2_FAIL, A3_FAIL, A4_FAIL), overall ROUTE_A_EXPLORATORY; Route B is disabled. We claim no arithmetic Euler factors, root number, target divisor, functional equation, automorphy, natural Hilbert–Pólya lift, or orbit-level injectivity within a sector. The literal firewall is NO_BAD_EULER_OR_ROOT_NUMBER.